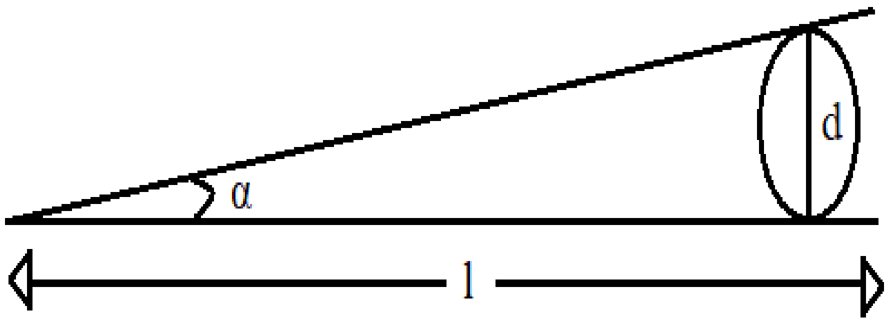
INTERFERENCE

Q1. Explain origin of colour in thin films?

If a film is illuminated by a beam of white light, different colours travel different lengths of path in the film in given time. Hence, different colours are reflected from different points of the film surface.

After passing through the lens some colour interferes constructively and are intensified and rest are fade out. Hence, film will appear to contain a distribution of the intensified colour.

Q2. Explain the determination of diameter of very thin wire or thickness of thin foil?



Here, a foil or wire is kept between two optically flat surfaces which are in contact with each other at another end.

Let ‘d’ be the diameter of the thin wire or the thickness of the thin foil. The angle of the wedge is α and λ is distance of wire from edge of wedge.

$$\tan \propto = \frac{d}{\square}$$

As α is very small, tanα=α

$$\propto = \frac{d}{\square}$$

Bandwidth, $\beta = \frac{\square}{2\mu\alpha}$

$$d = \frac{l}{2\mu\beta}$$

Where, λ→Wavelength of incident light

μ→Refractive index of film

β→ Bandwidth

d→ Thickness

Q3. Explain testing of surface flatness using interference?

The surface to be tested is kept under optically flat surface. Then according to the observation from travelling microscope the following things were found:

1]If the fringes are moving away from the centre then the surface is convex.

diagram

2]If the fringes are moving towards the centre then surface is concave.

diagram

3] If the surface is rough then the fringes appear irregular.

Diagram

Q4. Explain determination of refractive index of liquid?

In this experiment, few drops of given liquid can be introduced between lens and glass plate, and a thin liquid film is formed. By using microscope,

$$= \frac{(D_{n+m}^2 - D_n^2)_{air} \quad \mu}{4\,mR}$$

$$\mu_{air} = \frac{4\,mR}{(D_{n+m}^2 - D_n^2)}$$

$$(D_{n+m}^2 - D_n^2)_{liquid} = \frac{4\,mR}{\mu}$$

$$= \frac{\mu \ (D_{n+m}^2 - D_n^2)_{liquid}}{4\,Rm}$$

$$\mu_{liquid} = \frac{4\,Rm}{(D_{n+m}^2 - D_n^2)}$$

$$\frac{\mu_{liquid}}{\mu_{air}} = \frac{(D_{n+m}^2 - D_n^2)_{air}}{(D_{n+m}^2 - D_n^2)_{liquid}}$$

$$\mu_{liquid} = \frac{(D_{n+m}^2 - D_n^2)_{air}}{(D_{n+m}^2 - D_n^2)_{liquid}}$$

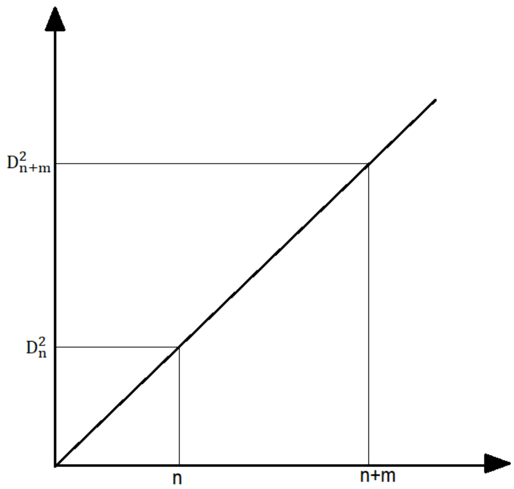
i.e. Newton ring shrinks with increase in refractive index of liquid.

Q5. How to determine wavelength of incident light?

Let R be the radius of curvature of lens in contact with the plate and be the wavelength of incident light. $D_{n+m} \wedge D_n$ are the diameter of (n+m) th and n th dark ring.

$$D_n^2 = \frac{4\,n\,R}{\mu}$$

$$D_{n+m}^2 = \frac{4(n+m)R}{\mu}$$



$$slope = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{D_{n+m}^2 - D_n^2}{n+m-n}$$

$$= \frac{D_{n+m}^2 - D_n^2}{m}$$

$$\frac{D_{n+m}^2 - D_n^2}{m} = \frac{4R}{\mu}$$

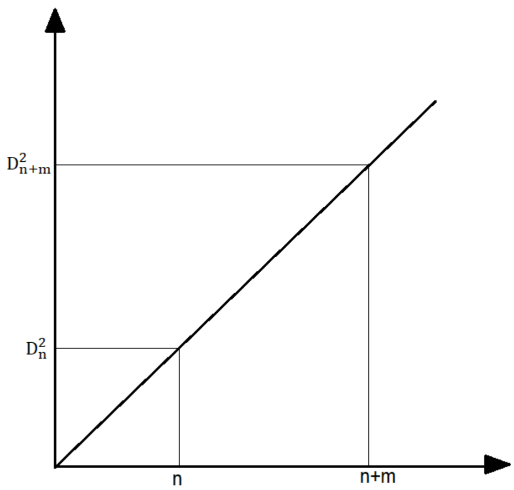
$$= \frac{D_{n+m}^2 - D_n^2}{m}) \frac{\mu}{4R}$$

Q6. How to determine wavelength of incident light?

Let R be the radius of curvature of lens in contact with the plate and be the wavelength of incident light. $D_{n+m} \wedge D_n$ are the diameter of (n+m) th and nth dark ring.

$$D_n^2 = \frac{4nR}{\mu}$$

$$D_{n+m}^2 = \frac{4(n+m)R}{\mu}$$



$$slope = \frac{y_2 - y_1}{x_2 - x_1}$$

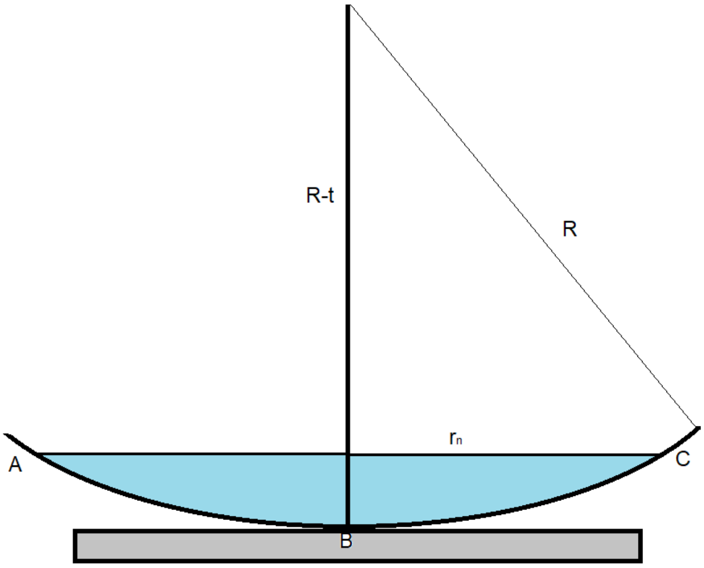
$$= \frac{D_{n+m}^2 - D_n^2}{n+m-n}$$

$$= \frac{D_{n+m}^2 - D_n^2}{m}$$

$$\frac{D_{n+m}^2 - D_n^2}{m} = \frac{4R}{\mu}$$

$$\left(\frac{D_{n+m}^2 - D_n^2}{m}\right)\frac{\mu}{4} = R$$

Q7. Derive newton's rings radius?



Consider a lens ABC kept on a plane surface. Here, R is radius of lens, r_n is radius of newtons rings and t is thickness.

:. By Pythagoras rule,

$$R^2 = (R-t)^2 + r_n^2$$

$$R^2 = R^2 + t^2 - 2Rt + r_n^2$$

t is very small, hence t^2 should be neglected,

$$2Rt = r_n^2$$

$$2t = \frac{rn^2}{R}$$

The path difference for reflected system,

$$\Delta = 2(\cos r)\ \mu t\ + \frac{❑}{2}$$

For air μ=1,

For normal incidence i=0, r=0

cos r= 1

$$\Delta = 2t \;+\; \frac{\square}{2}$$

$$\Delta = \frac{r_n^2}{R} + \frac{\square}{2}$$

<u>1] for Dark ring:</u>

$$\Delta = (2n+1)\frac{\square}{2}$$

$$\frac{r_n^2}{R} + \frac{\square}{2} = \frac{2n}{2} + \frac{\square}{2}$$

$$r_n^2 = \; nR$$

$$r_n = \sqrt{nR}$$

$$r_n \;\; \alpha \;\; \sqrt{n}$$

<u>2]for Bright ring:</u>

$$\Delta = n$$

$$\frac{r_n^2}{R} + \frac{\square}{2} = n$$

$$r_n^2 = \; R\left(\frac{2n\;\; -}{2}\right)$$

$$r_n = \sqrt{R\left(\frac{2n\;\; -}{2}\right)}$$

$$r_n = \sqrt{R\;\frac{\square}{2}\;(2n-1)}$$

$$r_n \;\; \alpha \;\; \sqrt{2n-1}$$